Chapter 3

Data Transmission

3-1 ANALOG AND DIGITAL

Data can be analog or digital.

***analog data refers to** information that is continuous(have an infinite number of values in a range);*

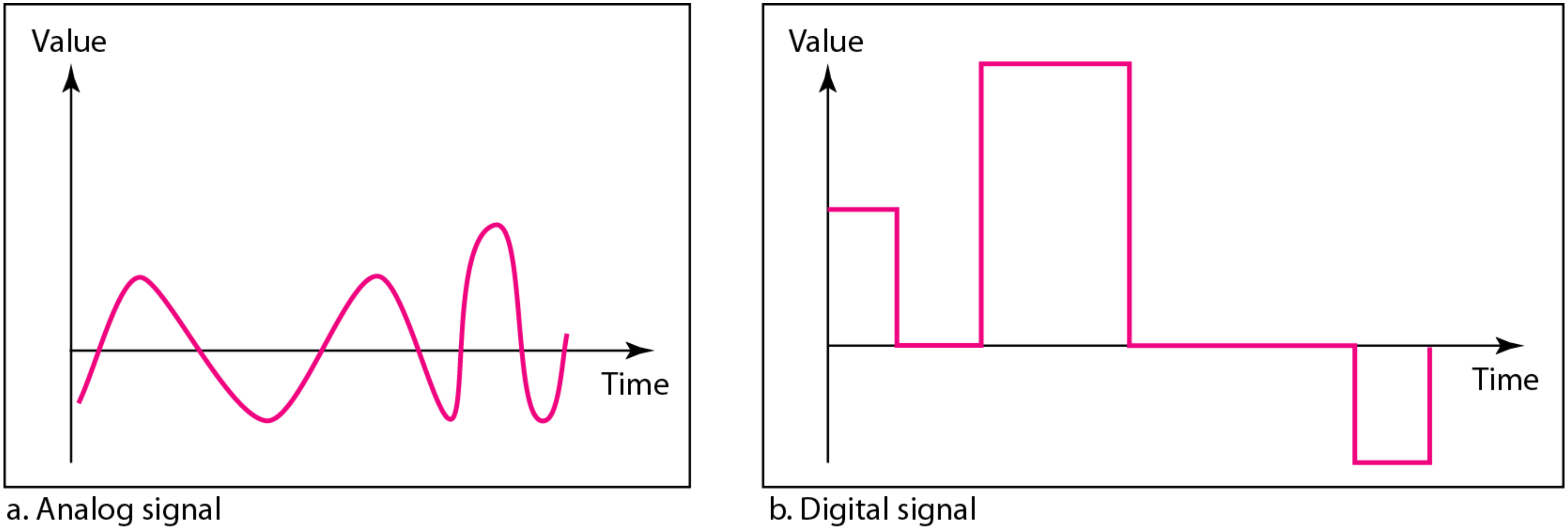
***digital data refers to** information that has discrete states (can have only a limited number of values).*

Analog data take on continuous values. Digital data take on discrete values.

Topics discussed in this section:

Analog and Digital Data
Analog and Digital Signal
Periodic and Nonperiodic Signal

Figure 3.1 *Comparison of analog and digital signals*



3-2 PERIODIC ANALOG SIGNALS

Periodic analog signals ***can be classified as simple or composite. A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals. A composite periodic analog signal that composed of multiple sine waves.***

Topics discussed in this section:

Sine Wave
Wavelength
Time and Frequency Domain
Composite Signals
Bandwidth

Sine Wave

- **Is the fundamental periodic signal**
- **Can be represented by three parameters**
  - **Peak amplitude (*A*)**
    - *Maximum value of the signal over time*
    - measured in volts
  - **Frequency (*f*)**
    - *Rate at which the signal repeats*
    - Hertz (Hz) or cycles per second
    - Period (*T*) :- *the amount of time for one repetition*
    - $T = 1/f$
  - **Phase (ϕ)**
    - *Relative position in time within a single period*
    - *describes the position of the waveform relative to time 0*

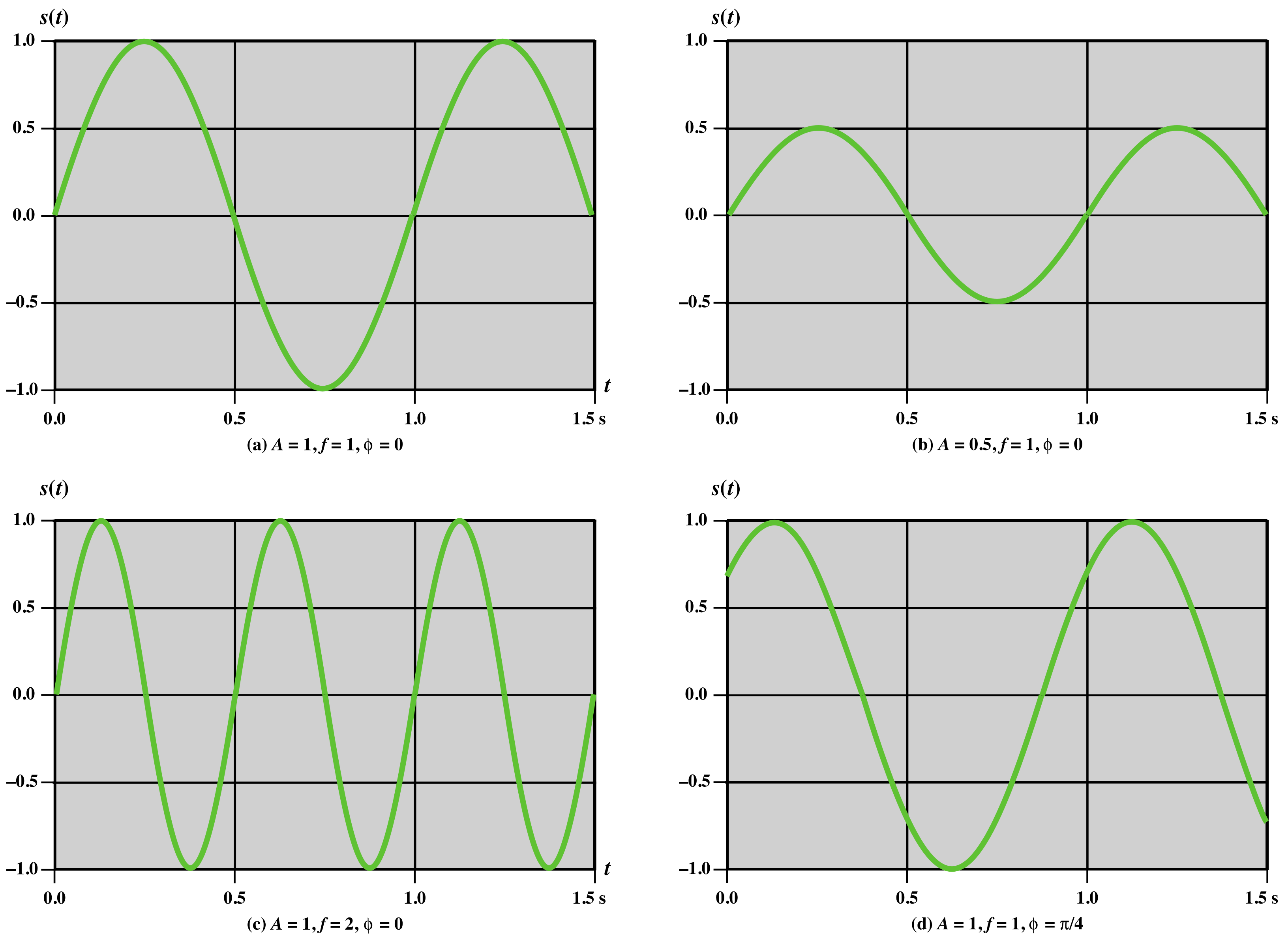
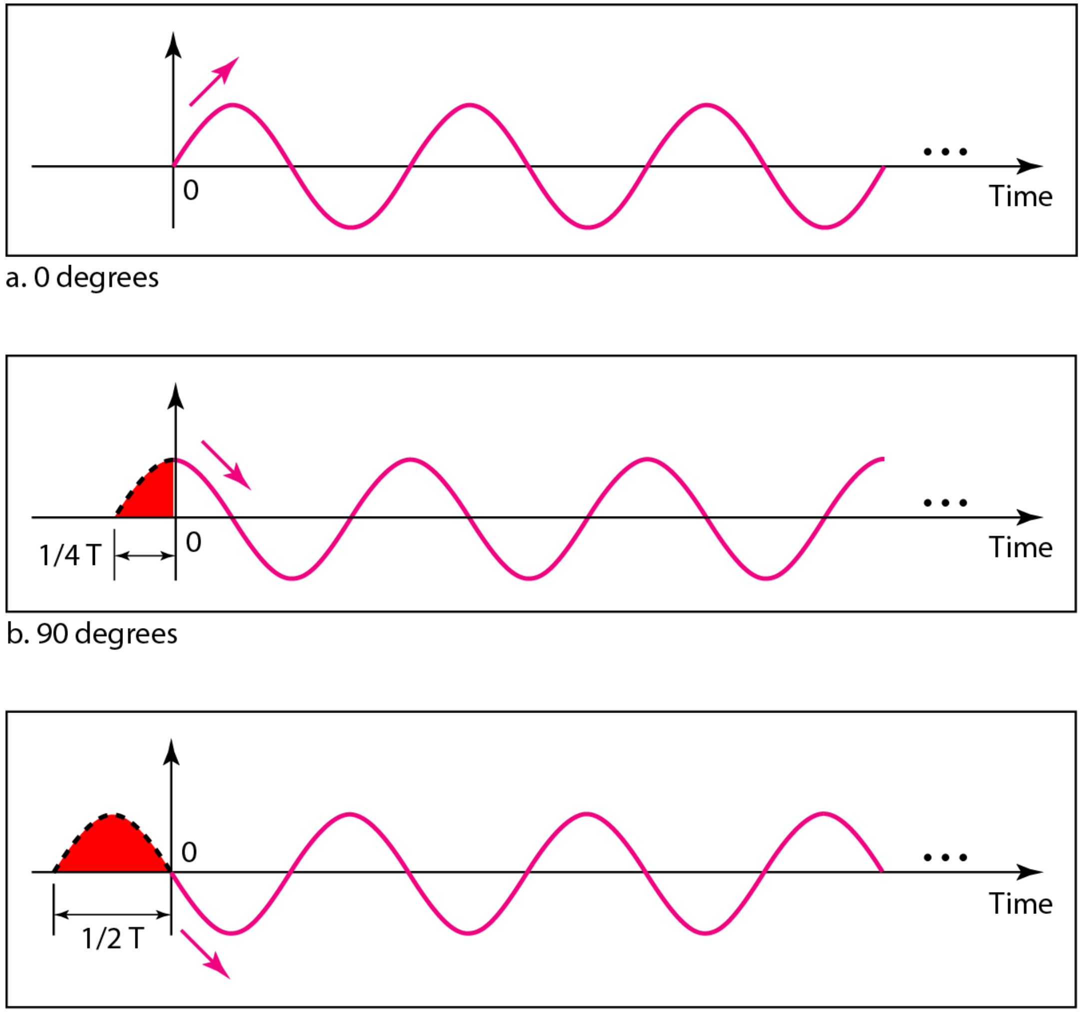


Figure 3.3 $s(t) = A \sin(2\pi ft + \phi)$

Note

Phase describes the position of the waveform relative to time 0.

Figure 3.5 *Three sine waves with the same amplitude and frequency, but different phases*

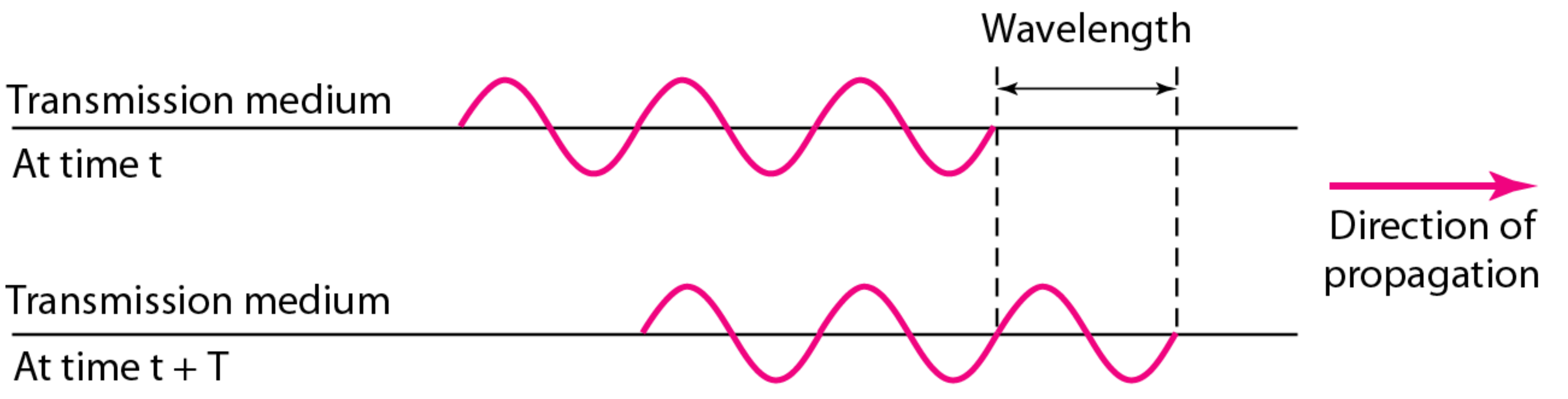


a. 0 degrees

b. 90 degrees

c. 180 degrees

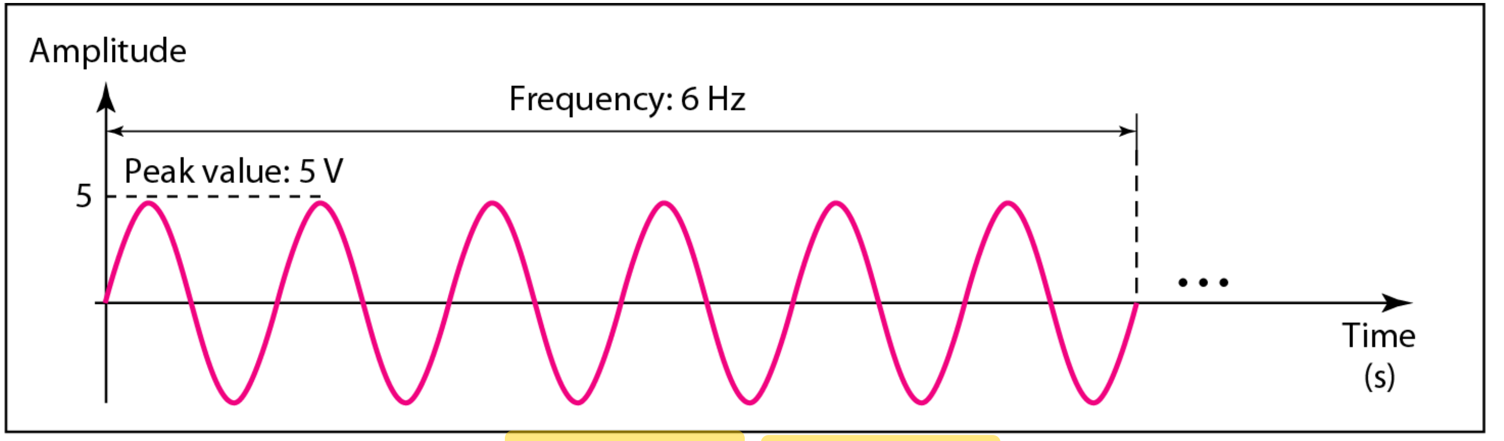
Figure 3.6 *Wavelength and period*



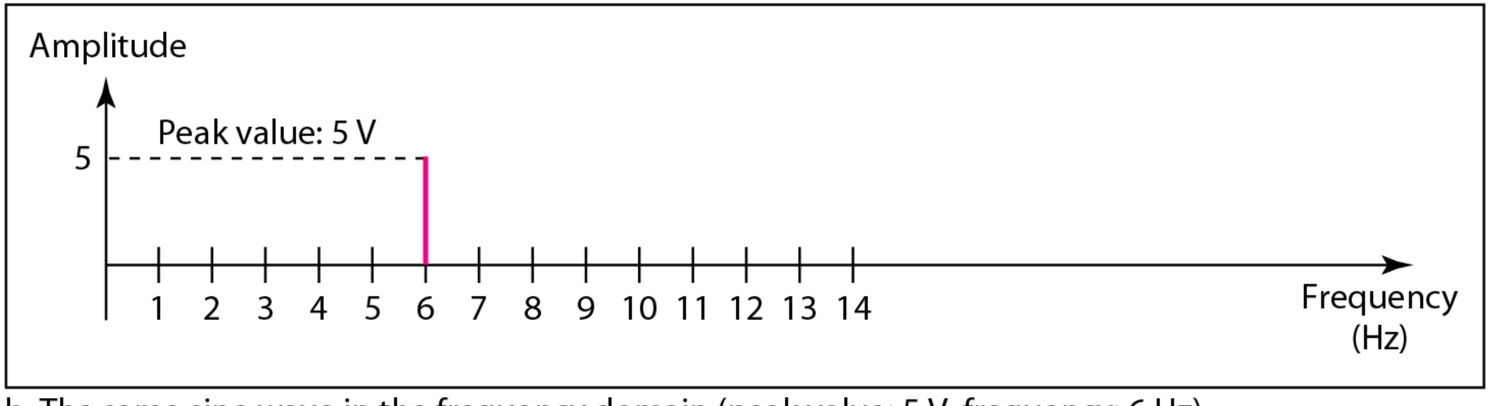
Note

A complete sine wave in the time domain can be represented by one single spike in the frequency domain.

Figure 3.7 *The time-domain and frequency-domain plots of a sine wave*

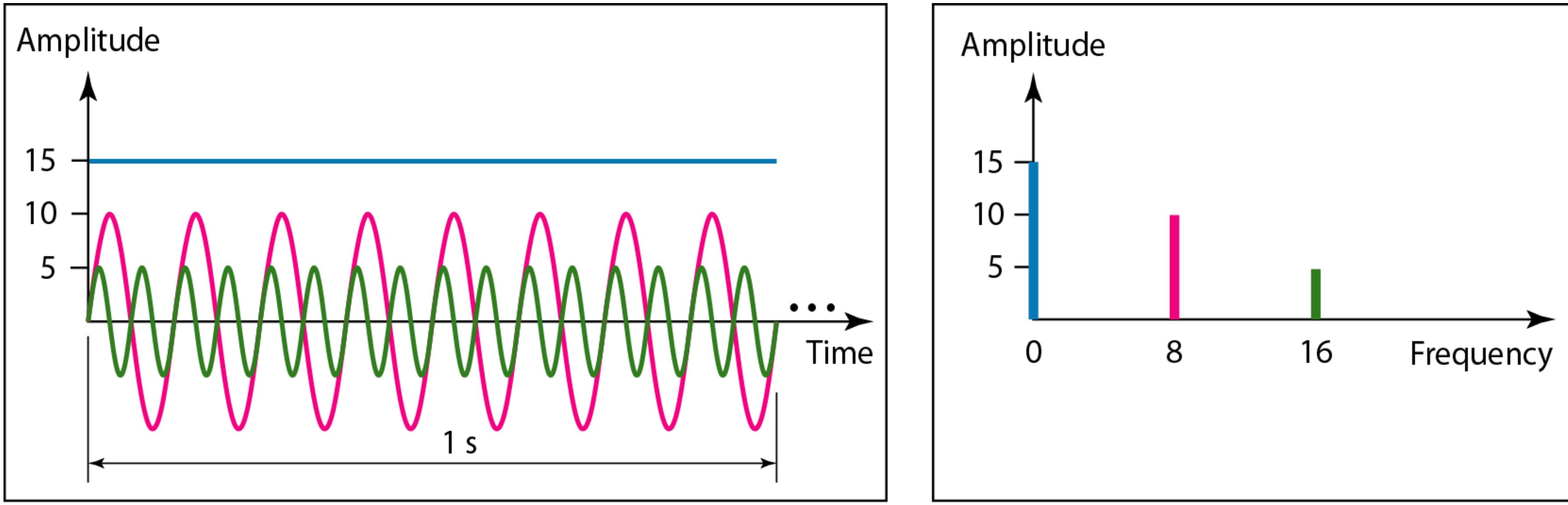


a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Figure 3.8 *The time domain and frequency domain of three sine waves*



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16

b. Frequency-domain representation of the same three signals

Composite Signals and Periodicity

- If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies.
- If the composite signal is nonperiodic, the decomposition gives a combination of sine waves with continuous frequencies.

Figure 3.9 *A composite periodic signal*

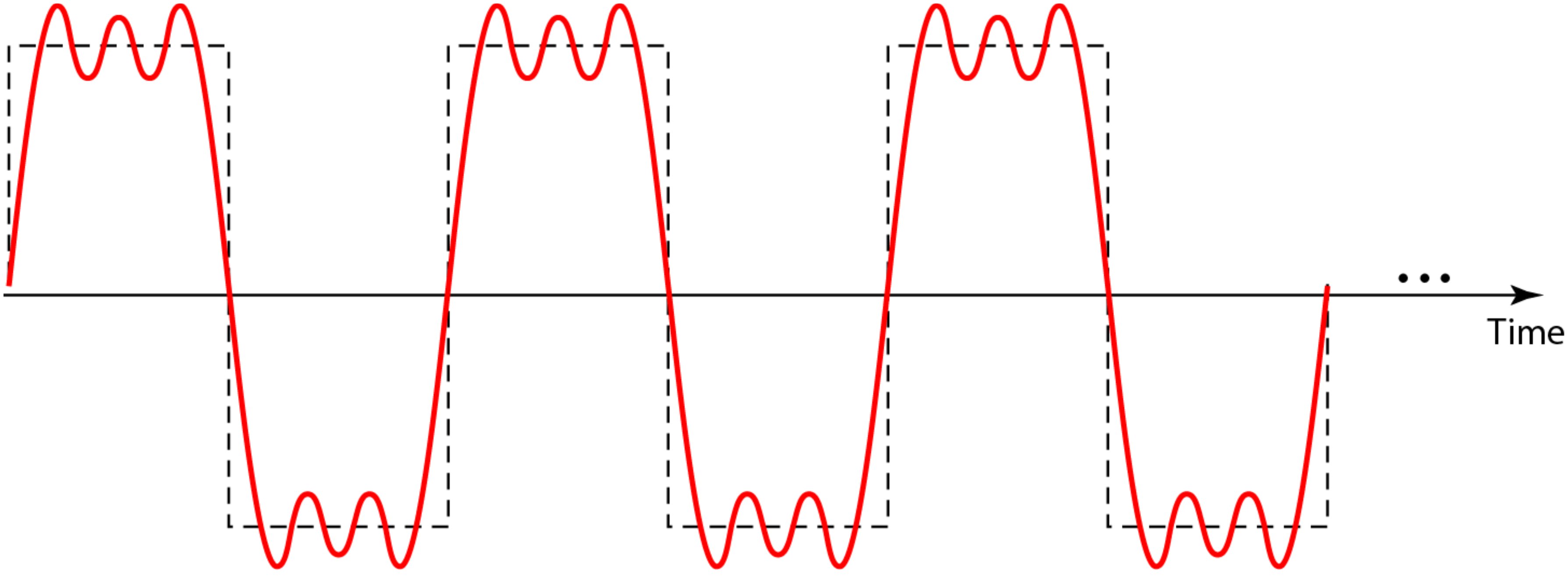
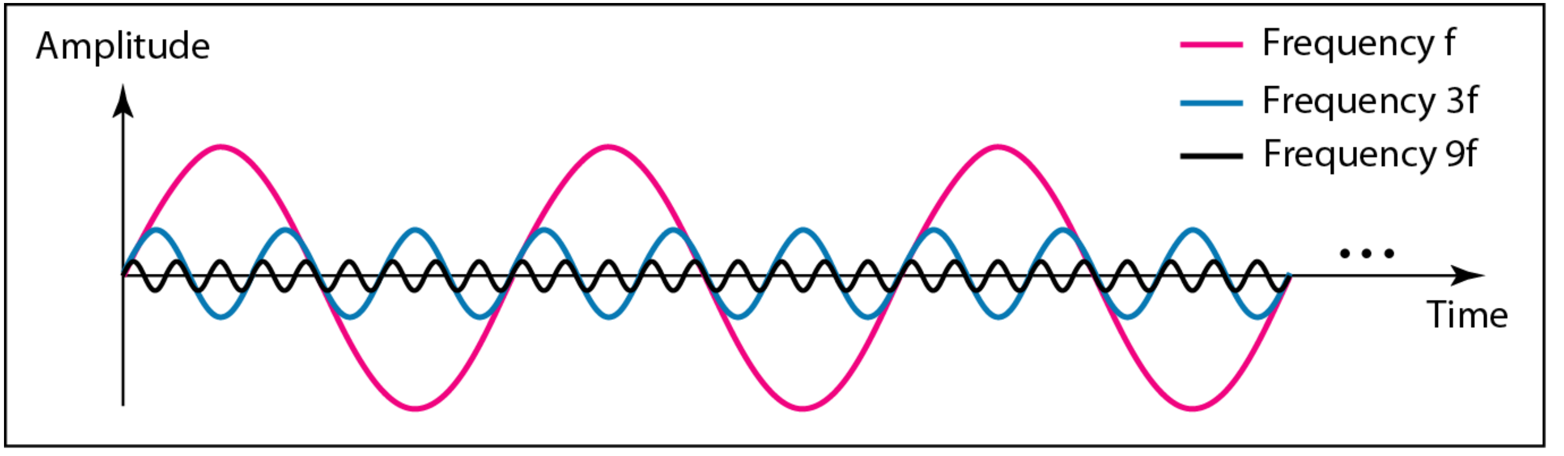
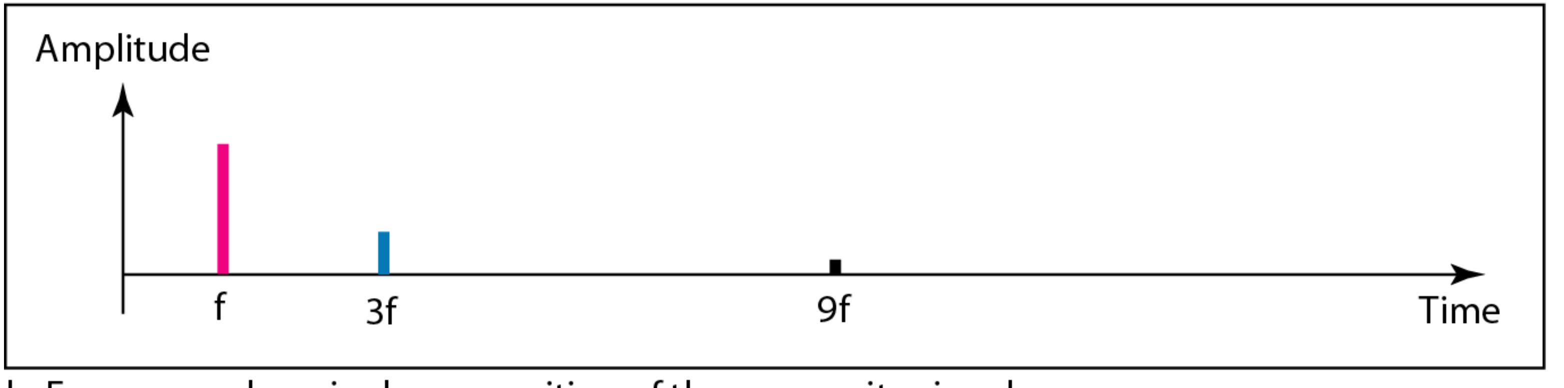


Figure 3.10 *Decomposition of a composite periodic signal in the time and frequency domains*

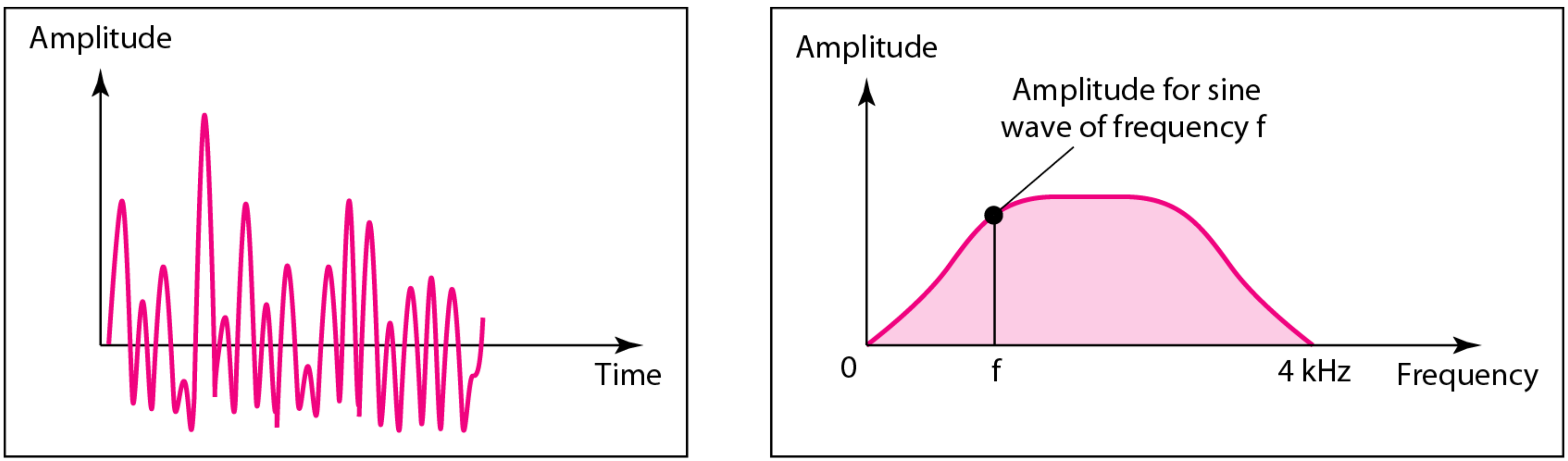


a. Time-domain decomposition of a composite signal



b. Frequency-domain decomposition of the composite signal

Figure 3.11 *The time and frequency domains of a nonperiodic signal*



a. Time domain

b. Frequency domain

Note

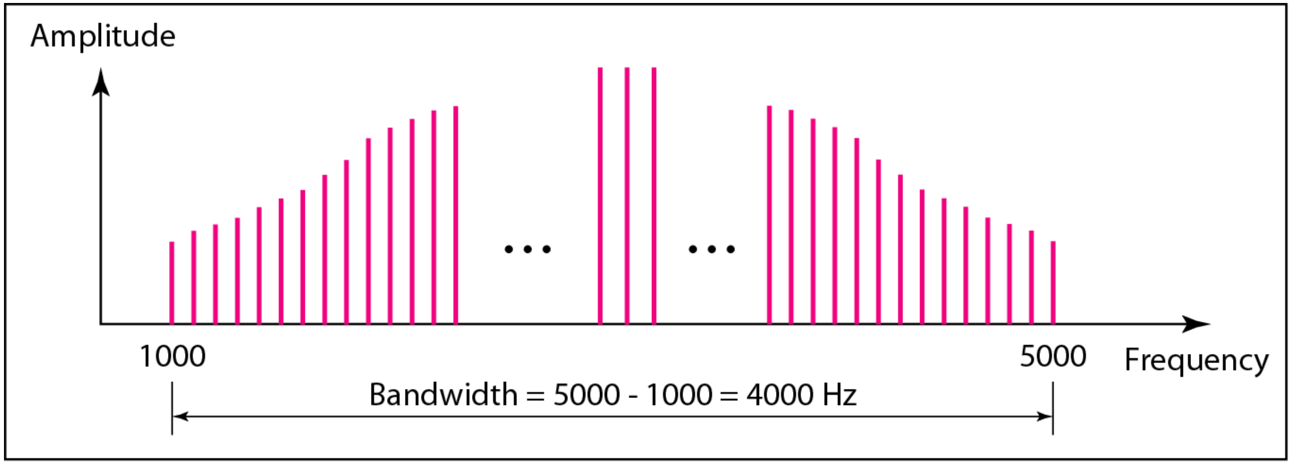
The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

Note

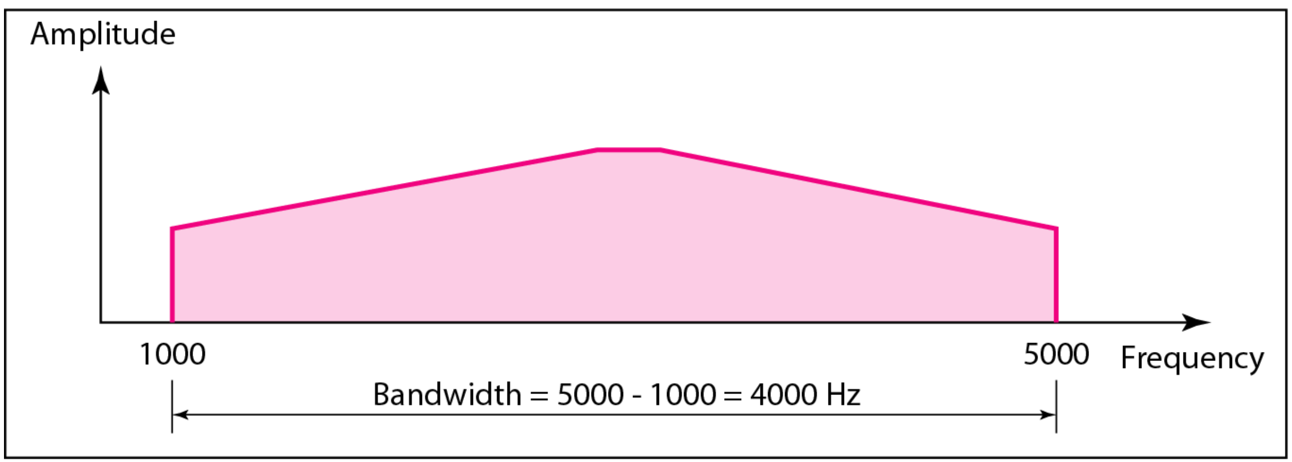
In networking, we use the term bandwidth in two contexts.

- The first, bandwidth in ***hertz***, refers to the *range of frequencies in a composite signal* or the range of frequencies that a channel can pass.
- The second, bandwidth in ***bits per second***, refers *to the speed of bit transmission in a channel or link* (***Capacity***)

Figure 3.12 *The bandwidth of periodic and nonperiodic composite signals*



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Example 3.10

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.

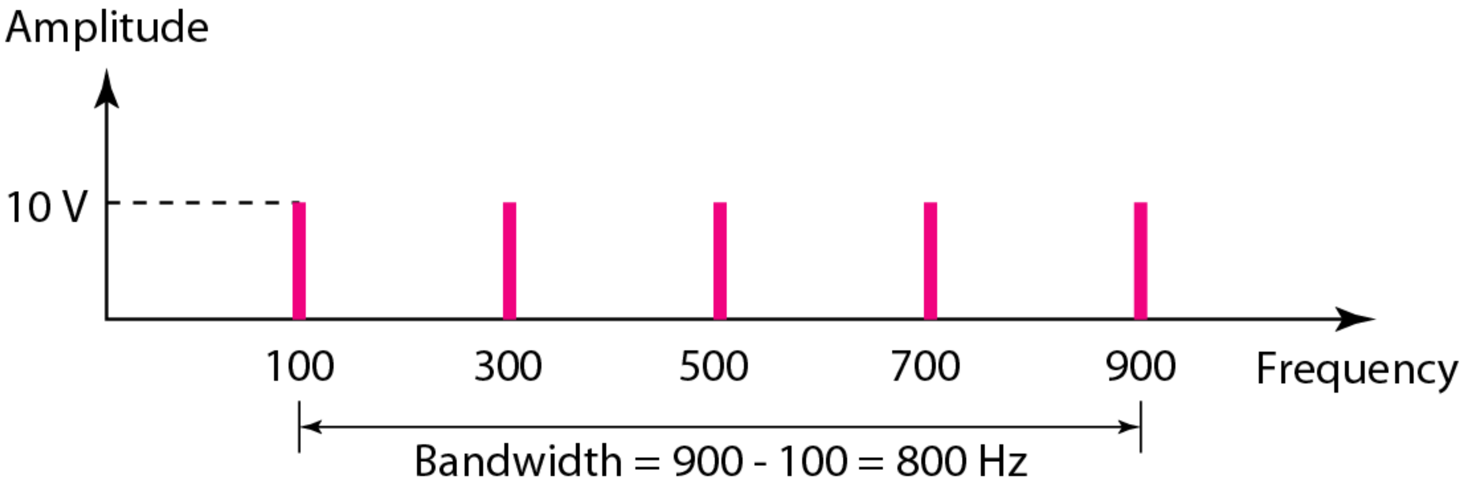
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$\boxed{B = f_h - f_l = 900 - 100 = 800 \text{ Hz}}$$

The spectrum has only five spikes, at 100, 300, 500, 700, and 900 Hz (see Figure 3.13).

Figure 3.13 *The bandwidth for Example 3.10*



Example 3.11

A periodic signal has a bandwidth of 20 Hz. The highest frequency is 60 Hz. What is the lowest frequency? Draw the spectrum if the signal contains all frequencies of the same amplitude.

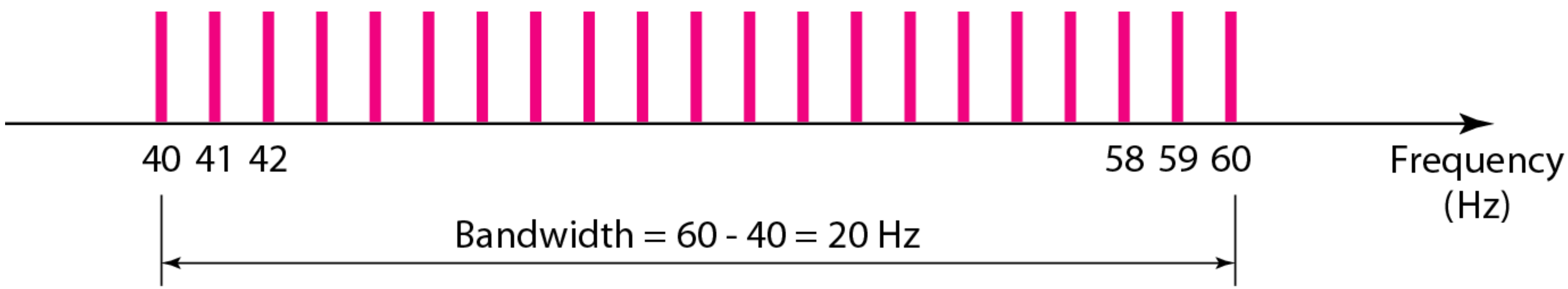
Solution

Let f_h be the highest frequency, f_l the lowest frequency, and B the bandwidth. Then

$$B = f_h - f_l \quad \Rightarrow \quad 20 = 60 - f_l \quad \Rightarrow \quad f_l = 60 - 20 = 40 \text{ Hz}$$

The spectrum contains all integer frequencies. ***We show this by a series of spikes (see Figure 3.14).***

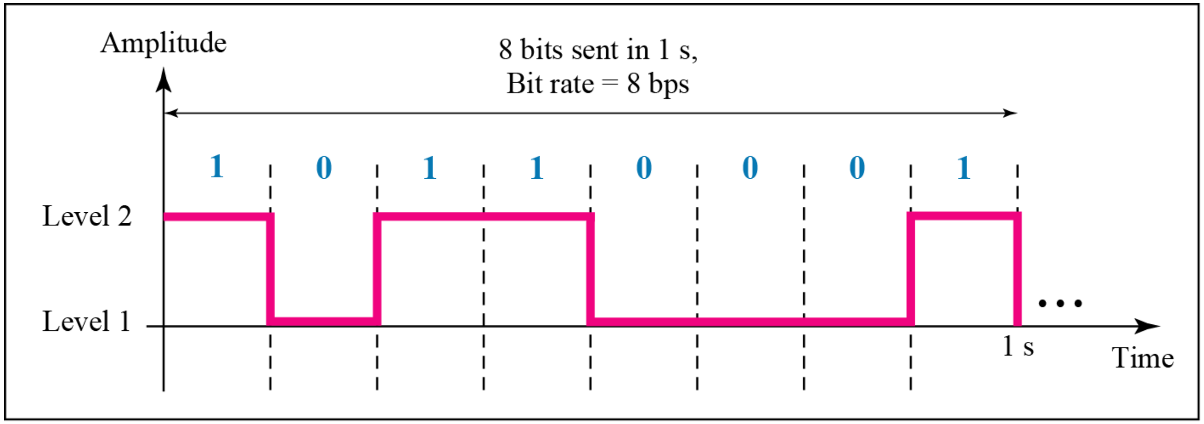
Figure 3.14 *The bandwidth for Example 3.11*



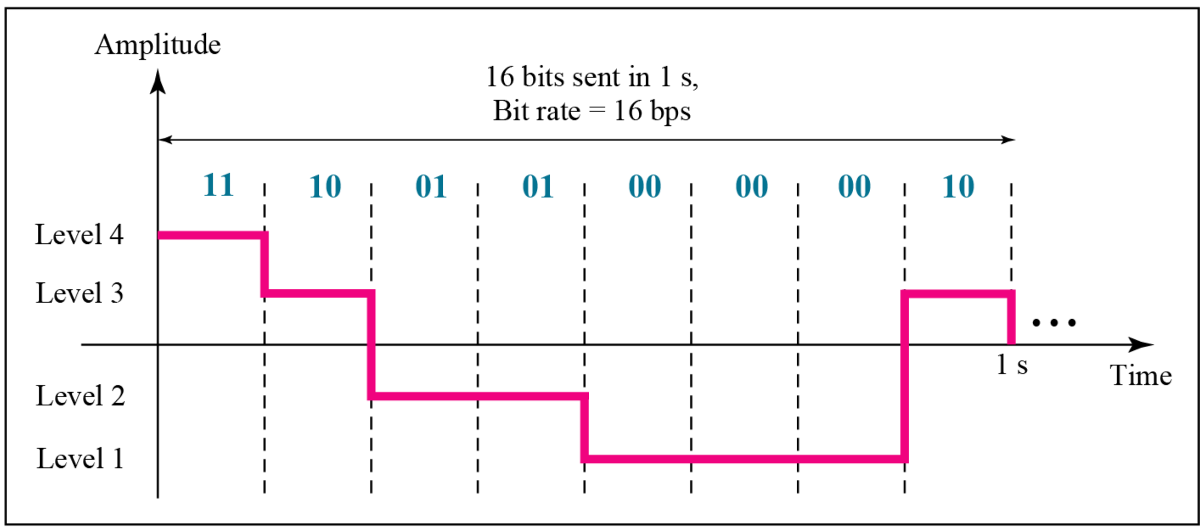
3-3 DIGITAL SIGNALS

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level.

Figure 3.16 *Two digital signals: one with two signal levels and the other with four signal levels*



a. A digital signal with two levels



b. A digital signal with four levels

Example 3.16

A digital signal has eight levels. How many bits are needed per level? We calculate the number of bits from the formula

$$\boxed{\text{Number of bits per level} = \log_2 8 = 3}$$

Number of levels = number of possibilities = 2 power n(bit)

Each signal level is represented by 3 bits.

Example 3.18

Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?

Solution

A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \times 24 \times 80 \times 8 = 1{,}636{,}000 \text{ bps} = 1.636 \text{ Mbps}$$

Example 3.19

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?

Solution

The bit rate can be calculated as

$$\boxed{2 \times 4000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}}$$

Example 3.20

What is the bit rate for high-definition TV (HDTV)?

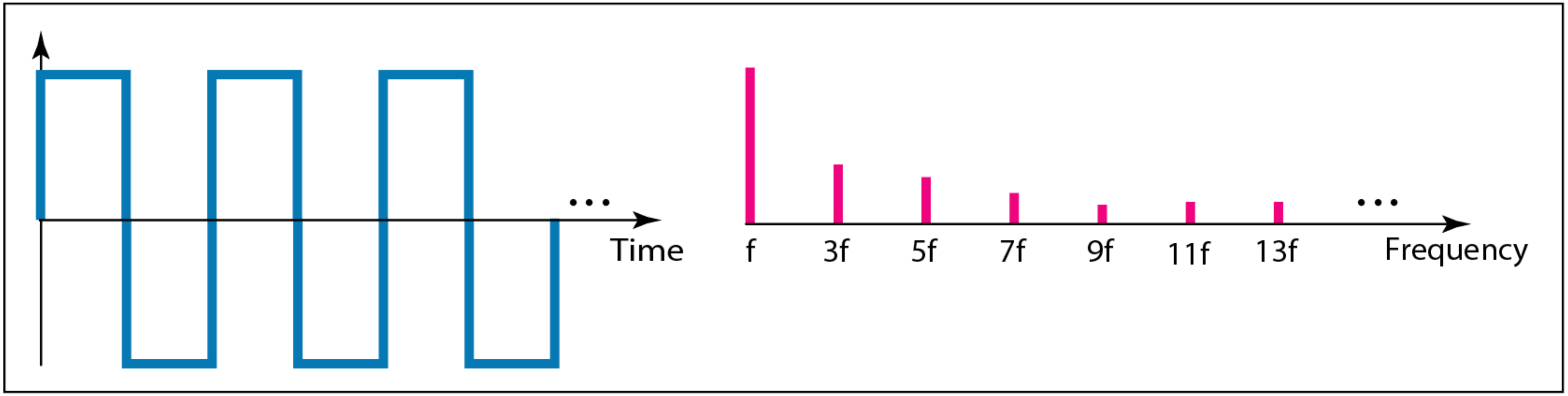
Solution

HDTV uses digital signals to broadcast high quality video signals. The HDTV screen is normally a ratio of 16 : 9. There are 1920 by 1080 pixels per screen, and the screen is renewed 30 times per second. Twenty-four bits represents one color pixel.

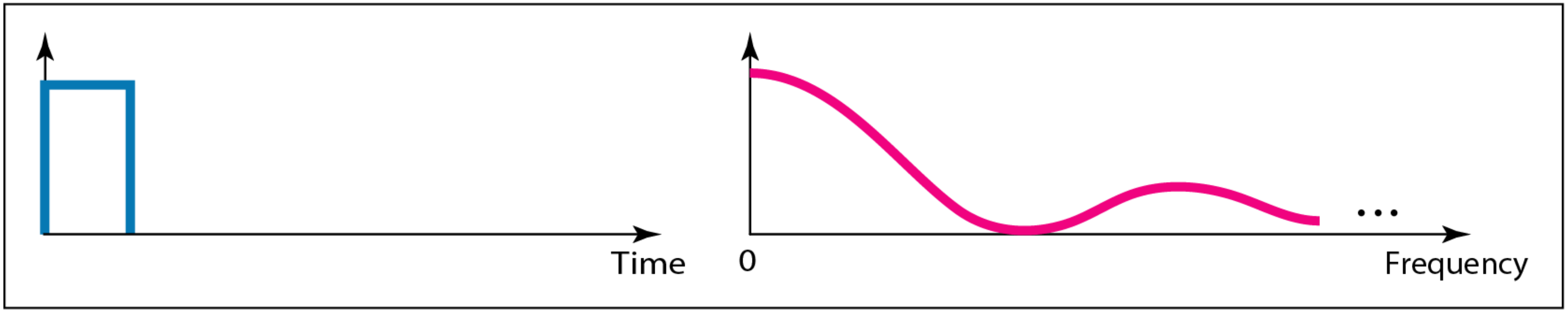
$$1920 \times 1080 \times 30 \times 24 = 1{,}492{,}992{,}000 \text{ or } 1.5 \text{ Gbps}$$

The TV stations reduce this rate to 20 to 40 Mbps through compression.

Figure 3.17 *The time and frequency domains of periodic and nonperiodic digital signals*

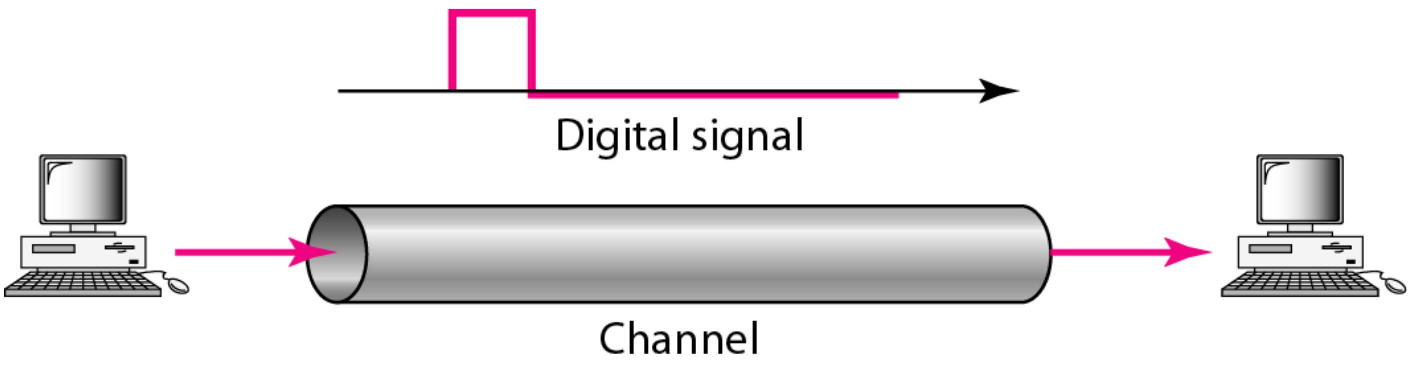


a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

Figure 3.18 *Baseband transmission*



Baseband vs. Broadband

Baseband:

- Baseband refers to *the original signal is directly transmitted without being modulated onto a carrier signal*.
- Examples of baseband transmission, Ethernet, HDMI, USB

Broadband:

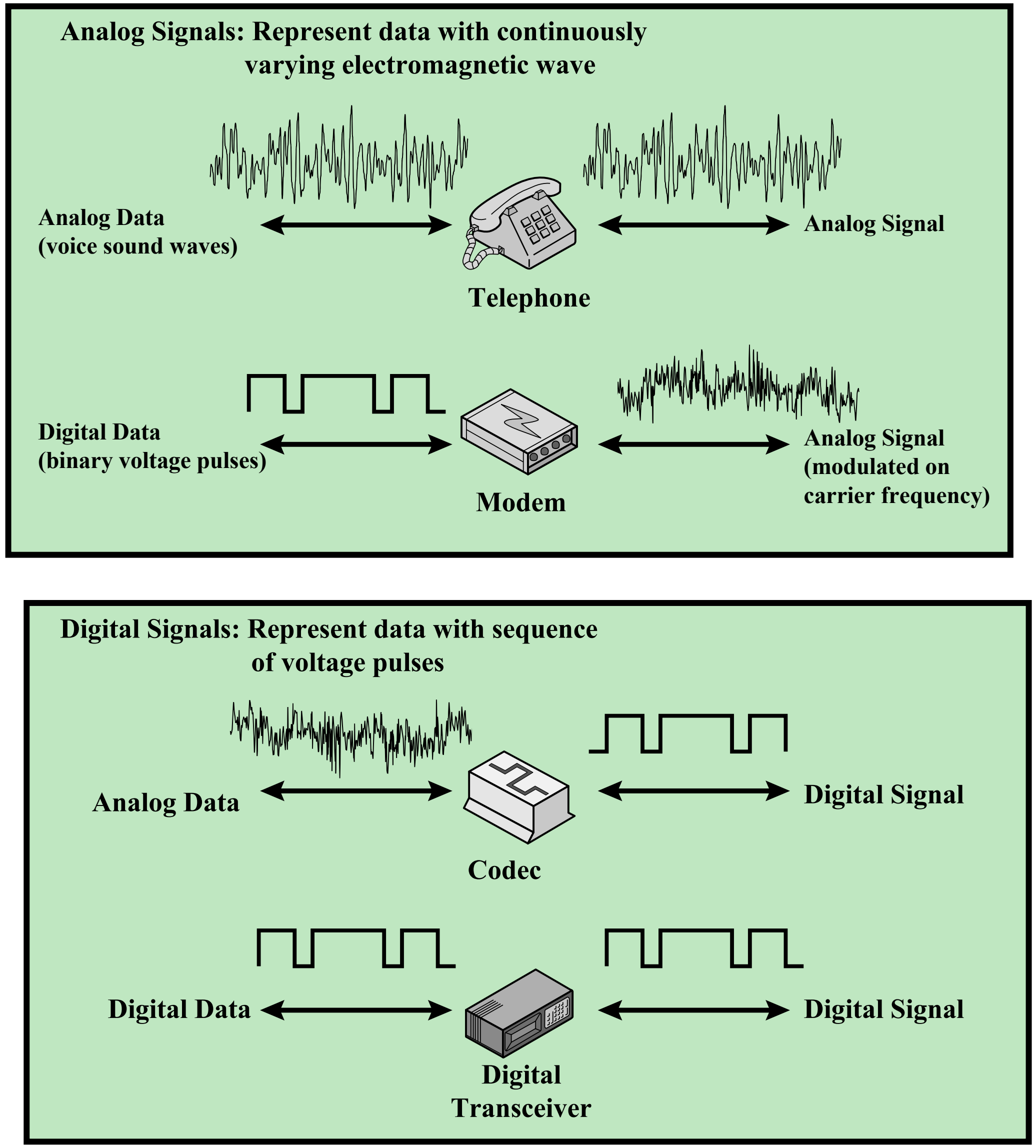
- Broadband refers to *a type of signaling or transmission in which the original signal is modulated onto a carrier signal,* **allowing for the transmission of multiple signals simultaneously** over a wide frequency range.
- Examples of broadband transmission: cable television, DSL, Wi-Fi.

Note

A digital signal is a composite analog signal with an infinite bandwidth.

Move to Digital

- Digital technology
  - VLSI technology has caused a continuing drop in the cost and size of digital circuits
- Data integrity
  - The use of repeaters has made it possible to transmit data longer distances over lower quality
- Capacity utilization
  - Satellite channels and optical fiber
- Security
  - Encryption techniques can be readily applied to digital data


Analog Signals: Represent data with continuously varying electromagnetic wave
Analog Data
(voice sound waves)
Telephone
Analog Signal
Digital Data
(binary voltage pulses)
Modem
Analog Signal
(modulated on carrier frequency)
Digital Signals: Represent data with sequence of voltage pulses
Analog Data
Codec
Digital Signal
Digital Data
Digital Transceiver
Digital Signal

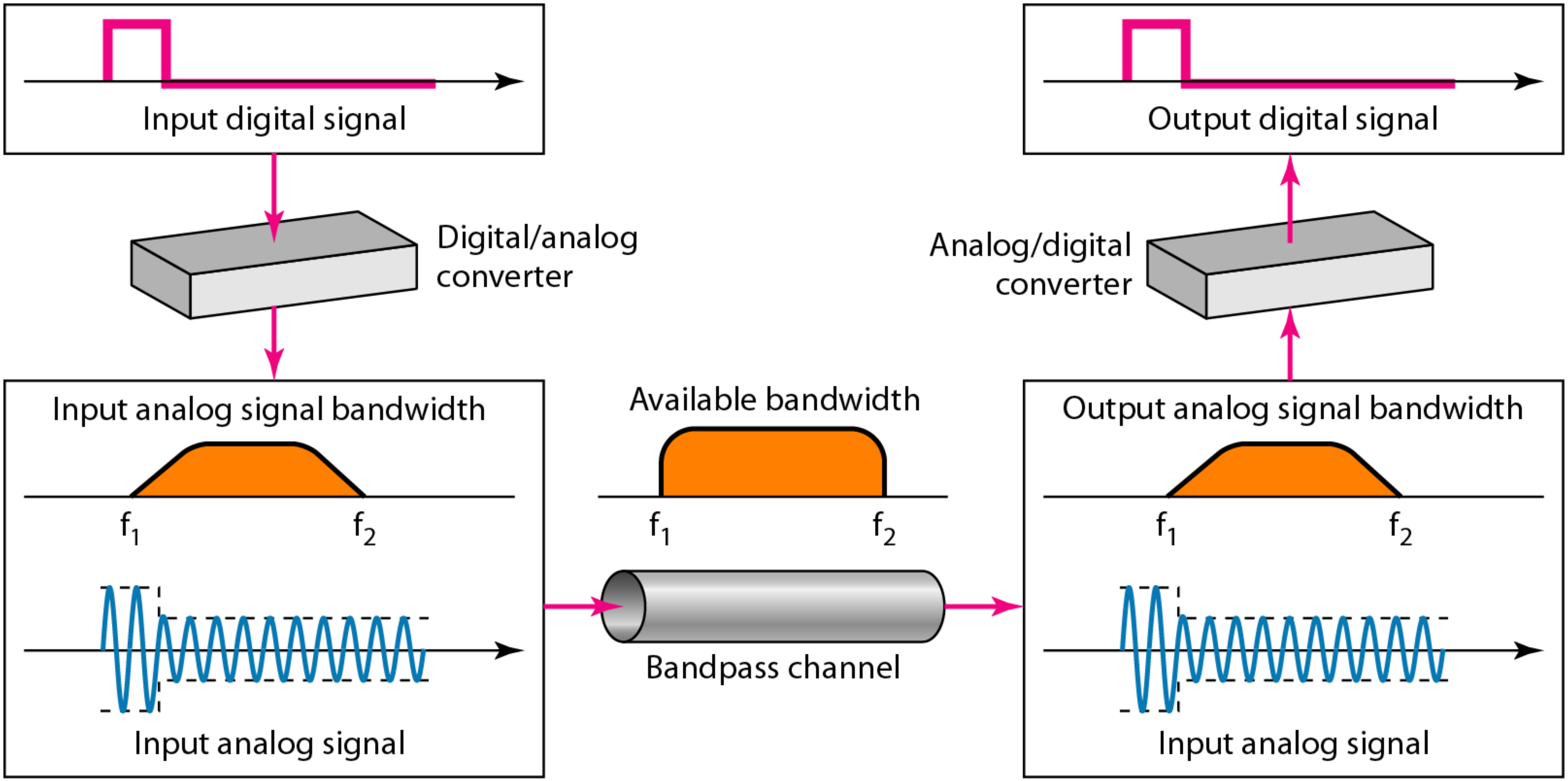
Modulation

Modulation in communication is *the process of changing one or more properties of a carrier signal*,(amplitude,frequency,phase) *to encode information*. The modulated carrier signal is then transmitted over a communication channel, such as a copper wire, fiber optic cable, or radio wave.

The different types of modulation include:

- Amplitude modulation (AM)
- Frequency modulation (FM)
- Phase modulation (PM)

Figure 3.24 *Modulation of a digital signal for transmission on a bandpass channel*



3-4 TRANSMISSION IMPAIRMENT

- *Signals travel through transmission media*
- *transmission media is not perfect.*
- *The imperfection causes signal impairment.*

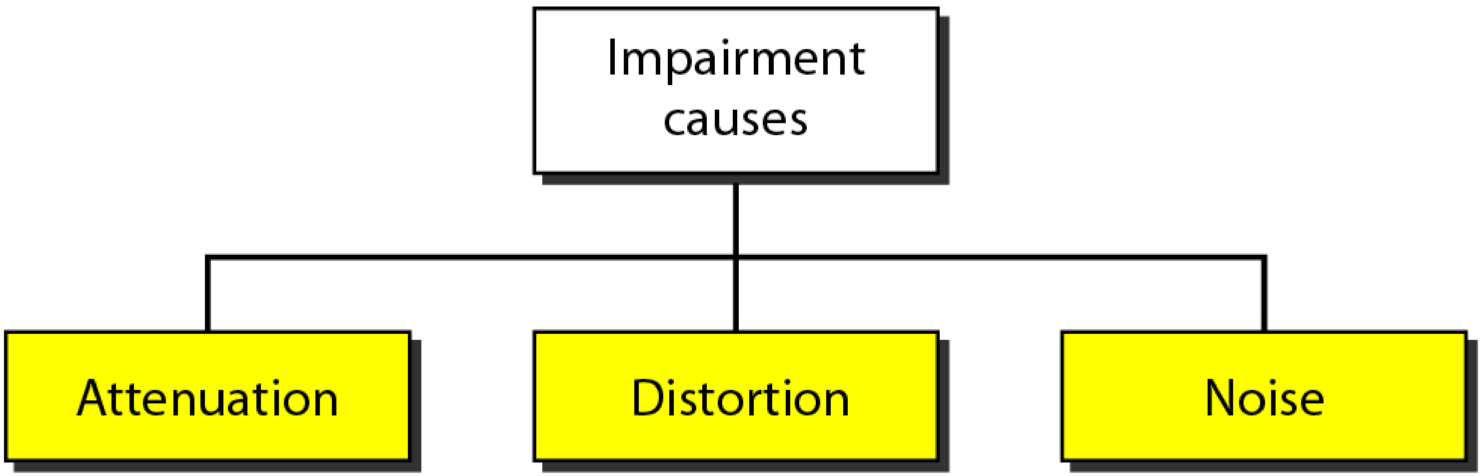
This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

Signal received may differ from signal transmitted causing:

Analog :- degradation of signal quality

Digital :- bit errors

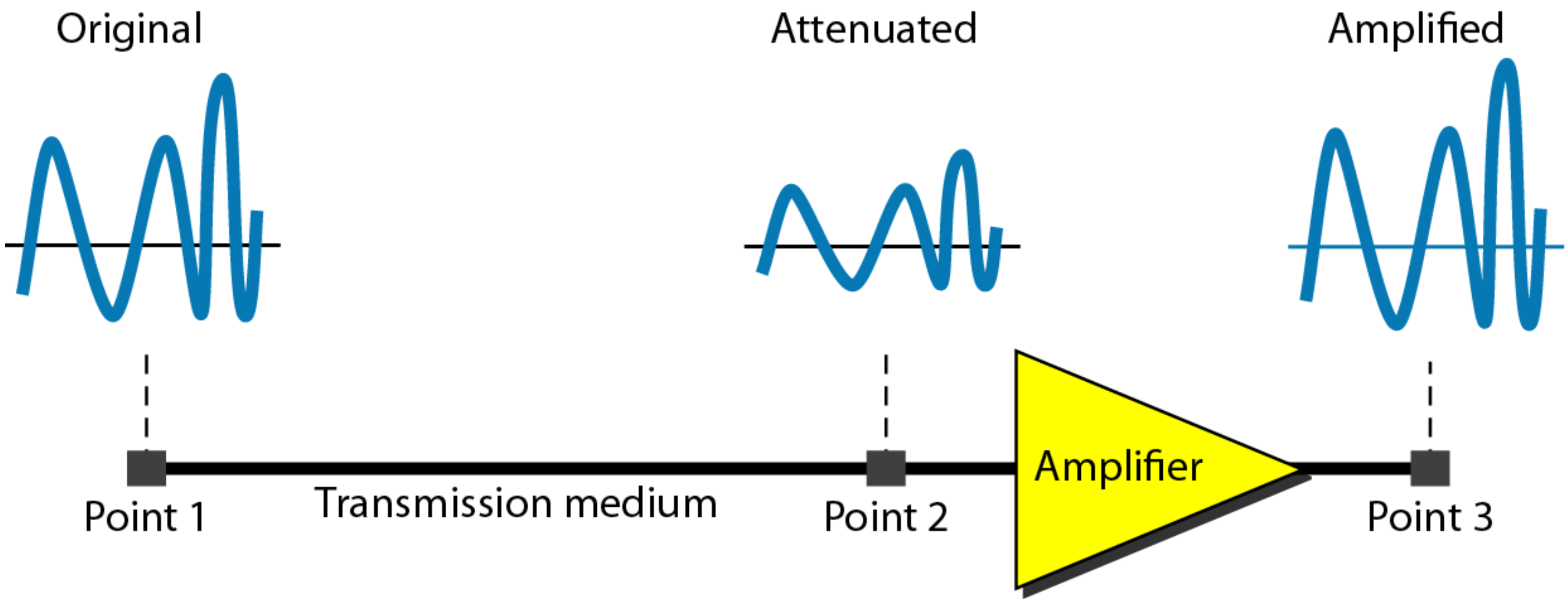
Figure 3.25 *Causes of impairment*



Attenuation in(DB): $\boxed{10 \log_{10} \frac{P_2}{P_1}}$

- *Signal strength falls off with distance over any transmission medium (الباور سواء زيادة او نقصان)*

- *Amplifiers are used* to compensate for this loss of energy by amplifying the signal.



Example 3.26

Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P_2 is (1/2)P_1. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

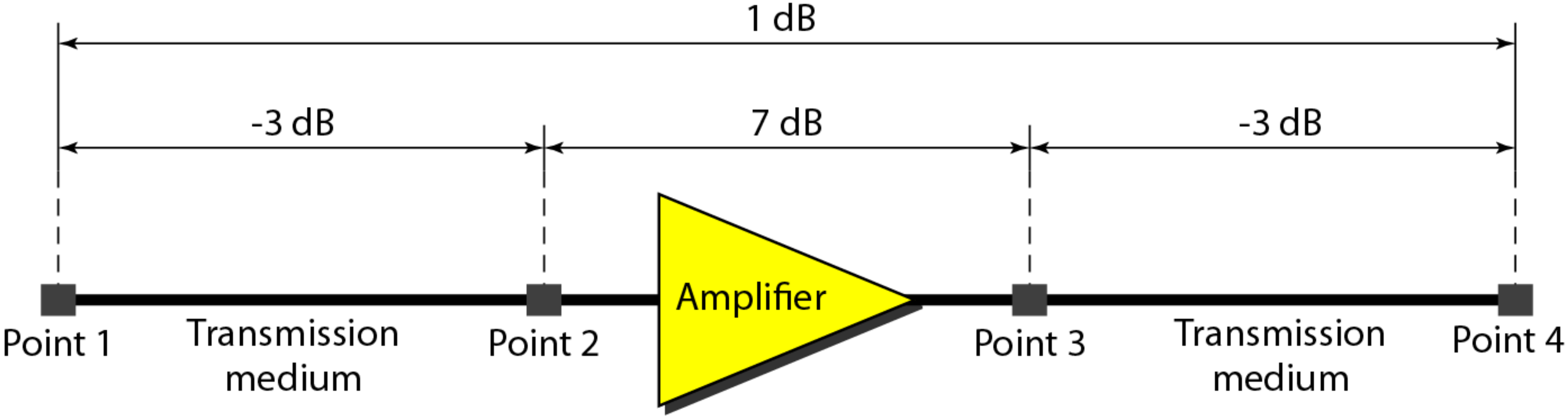
Example 3.27

A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1}$$

$$= 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Figure 3.27 *Decibels for Example 3.28*



Example 3.28

One *reason that engineers use the decibel* ***to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two. In Figure 3.27 a signal travels from point 1 to point 4. In this case, the decibel value can be calculated as***

$$\boxed{\mathrm{dB} = -3 + 7 - 3 = +1}$$

Example 3.30

The loss in a cable is usually defined in decibels per kilometer (dB/km). If the signal at the beginning of a cable with −0.3 dB/km has a power of 2 mW, what is the power of the signal at 5 km?

Solution

The loss in the cable in decibels is 5 × (−0.3) = −1.5 dB. We can calculate the power as

$$dB = 10 \log_{10} \frac{P_2}{P_1} = -1.5$$

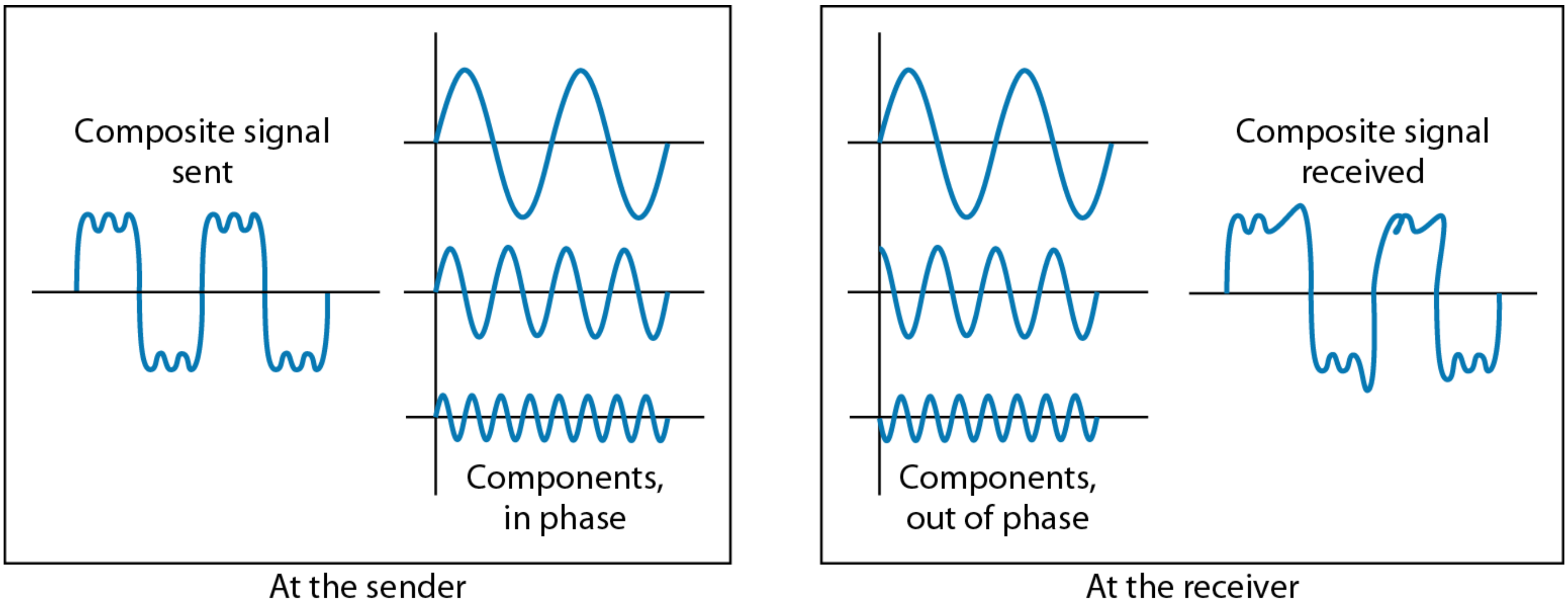
$$\frac{P_2}{P_1} = 10^{-0.15} = 0.71$$

$$P_2 = 0.71 P_1 = 0.7 \times 2 = 1.4 \text{ mW}$$

Delay Distortion

- Means that the signal changes its form or shape
- Occurs in transmission cables such as twisted pair, coaxial cable, and optical fiber
  - Does not occur when signals are transmitted through the air by means of antennas
- Various frequency components arrive at different times resulting in phase shifts between the frequencies

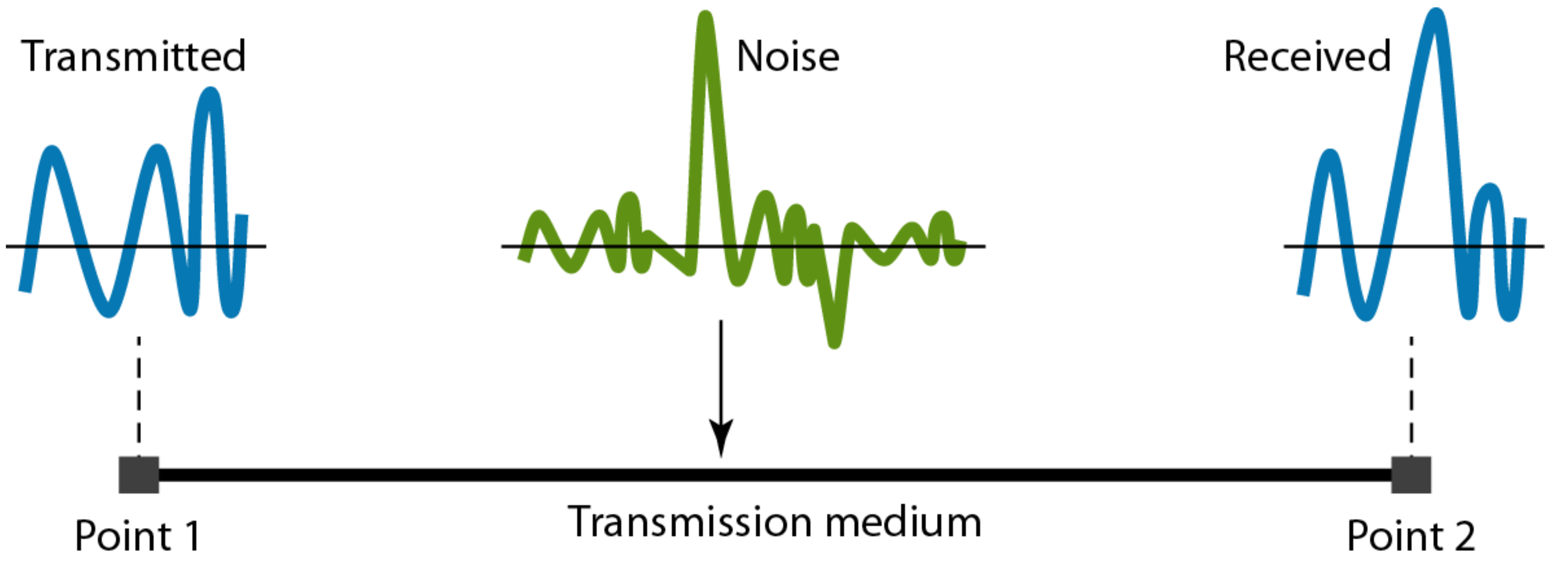
Figure 3.28 *Distortion*



Noise

- Unwanted signals inserted between transmitter and receiver
- There are different types of noise
  - Thermal - random noise of electrons in the wire creates an extra signal
  - Induced - from motors and appliances, devices act are transmitter antenna and medium as receiving **antenna**.
  - Crosstalk - same as above but between two wires.
  - Impulse - Spikes that result from power lines, lightning, etc.

Figure 3.29 *Noise*



Signal to Noise Ratio (SNR)

- *To measure the quality of a system* the SNR is often used. It indicates the strength of the signal wrt the noise power in the system.
- SNR =(signal_POWER / noise_POWER)
- IN DB = 10log(SNR)
- It is usually given in dB and referred to as SNR_{dB}.

Example 3.31

The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNR_{dB} ?

Solution

The values of SNR and SNR_{dB} can be calculated as follows:

$$SNR = \frac{10{,}000\ \mu W}{1\ mW} = 10{,}000$$
$$SNR_{dB} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

Example 3.32

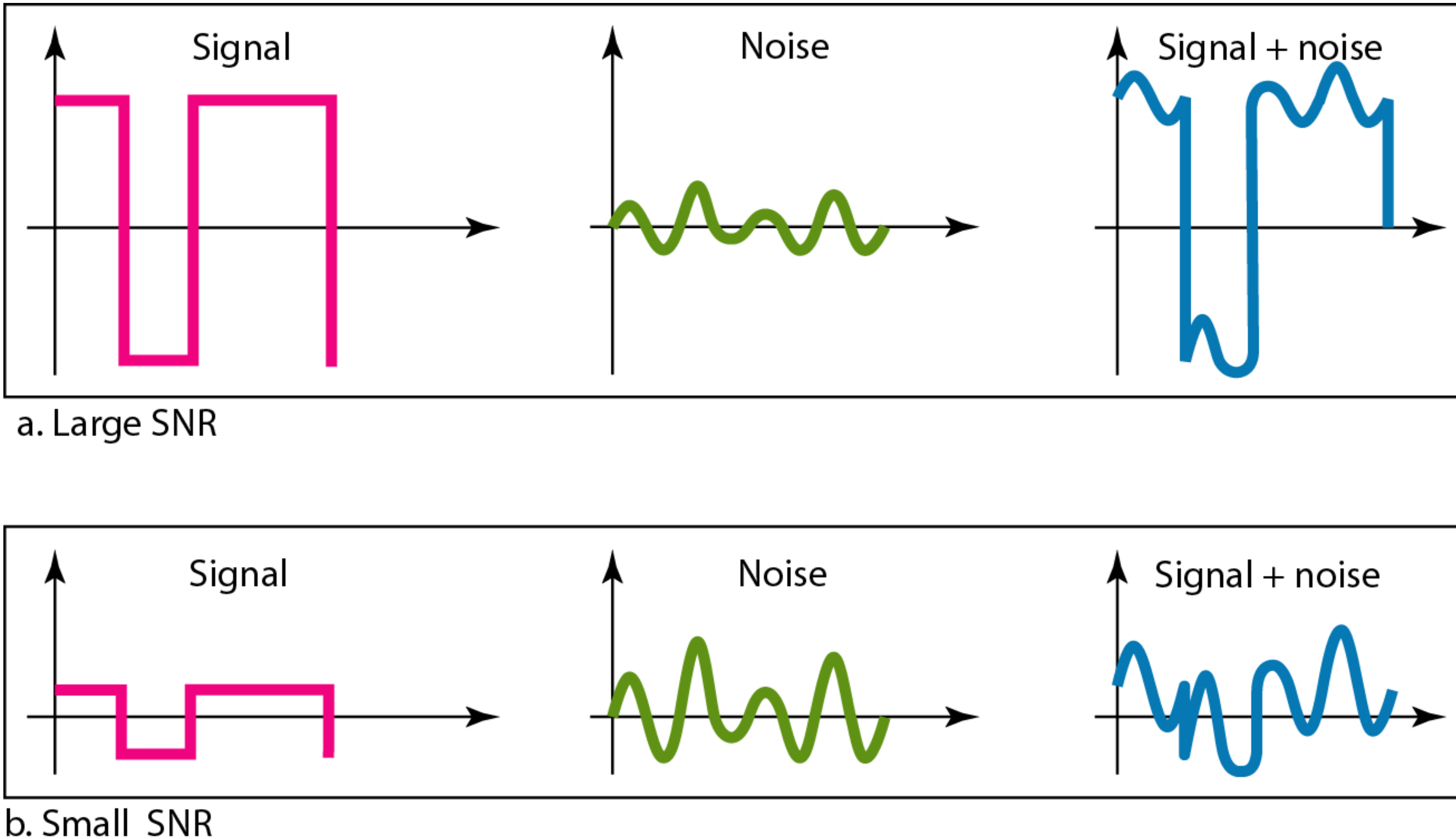
The values of SNR and SNR_{dB} for a noiseless channel are

$$SNR = \frac{\text{signal power}}{0} = \infty$$

$$SNR_{dB} = 10 \log_{10} \infty = \infty$$

We can never achieve this ratio in real life; it is an ideal.

Figure 3.30 *Two cases of SNR: a high SNR and a low SNR*



3-5 DATA RATE LIMITS

How fast we can send data, in bits per second, over a channel

Data rate depends on three factors(Data Rate Limits):

1. *The bandwidth available*
2. *The level of the signals we use*
3. *The quality of the channel (the level of noise)*

Topics discussed in this section:

Noiseless Channel: Nyquist Bit Rate
Noisy Channel: Shannon Capacity
Using Both Limits

Note

Increasing the levels of a signal may reduce the reliability of the system.

Channel Capacity

Maximum rate at which data can be transmitted over a given communications channel under given conditions

Data rate	Bandwidth	Noise	Error rate
in bits per second (bps) at which data can be communicated	in cycles per second, or hertz Constrained by transmitter and medium	The average level of noise over the communications path	The rate at which errors occur

Nyquist Theorem

- Nyquist gives the upper bound for the bit rate of a transmission system by calculating the bit rate directly from the number of bits in a symbol (or signal levels) and the bandwidth of the system (assuming 2 symbols/per cycle and first harmonic).
- Nyquist theorem states that for a noiseless channel:

$$\boxed{C = 2\,B\,\log_2 2^{n(\#BIT)} \quad \#LEVEL}$$

C= capacity in bps

B = bandwidth in Hz

Example 3.34

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\boxed{\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}}$$

Example 3.35

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Example 3.36

capacity

We need to send 265 kbps over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need?

Solution

We can use the Nyquist formula as shown:

$$265{,}000 = 2 \times 20{,}000 \times \log_2 L$$

$$\log_2 L = 6.625 \qquad L = 2^{6.625} = 98.7 \text{ levels}$$

Since this result is not a power of 2, we need to either:

- ***increase the number of levels If we have 128 levels, the bit rate is 280 kbps.*** (اقرب رقم يديك 2 اس)
- ***reduce the bit rate. If we have 64 levels, the bit rate is 240 kbps.***

Shannon Capacity Formula

- Considering the relation of data rate, noise and error rate
- Shannon developed formula relating these to signal to noise ratio (in decibels)

- SNR_{db}=10 $\log_{10}$ (signal/noise)
- Capacity $C = B \log_2(1+SNR)$
  - Theoretical maximum capacity
  - Get much lower rates in practice

Shannon's Theorem

- Shannon's theorem gives the capacity of a system in the presence of noise.

$$C = B \log 2(1 + SNR)$$

$$C = B \times \frac{SNR_{dB}}{3}$$

Example 3.37

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$\boxed{C = B \log_2 (1 + \mathrm{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0}$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example 3.38

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163$$
$$= 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example 3.39

The signal-to-noise ratio is often given in decibels. Assume that $SNR_{dB} = 36$ and the channel bandwidth is 2 MHz. The theoretical channel capacity can be calculated as

$$SNR_{dB} = 10 \log_{10} SNR \rightarrow SNR = 10^{SNR_{dB}/10} \rightarrow SNR = 10^{3.6} = 3981$$

$$C = B \log_2 (1 + SNR) = 2 \times 10^6 \times \log_2 3982 = 24 \text{ Mbps}$$

Example 3.40

For practical purposes, when the SNR is very high, we can assume that SNR + 1 is almost the same as SNR. In these cases, the theoretical channel capacity can be simplified to

$$\boxed{C = B \times \frac{\text{SNR}_{\text{dB}}}{3}}$$

For example, we can calculate the theoretical capacity of the previous example as

$$\boxed{C = 2\text{ MHz} \times \frac{36}{3} = 24\text{ Mbps}}$$

Example 3.41

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

- **First, we use the Shannon formula to find the upper limit.**

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

Example 3.41 (continued)

The Shannon formula gives us 6 Mbps, the upper limit.

- **For better performance we choose something lower**, ***4Mbps***,
- ***Then we use the Nyquist formula to find the number of signal levels.***

$$\boxed{4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \Longrightarrow \quad L = 4}$$

Note

The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.